

Macroscopic Pattern Joint Transition Probabilities in Hopfield Networks and Dense Associative Memory

Theoretical Physics & Neural Computation Note

August 2026

Abstract

This paper presents the exact mathematical formulation of macroscopic pattern-to-pattern transition probabilities in sequential associative memories. We analyze handwritten matrix projection dynamics, derive the joint conditional probability factorization from microscopic parallel Glauber spin dynamics, and demonstrate its extension to continuous Modern Hopfield Networks and Dense Associative Memory (DAM).

1 Interpretation of Handwritten Notes

The general bilinear matrix projection rule defines how state-transition kernels are embedded into neural coupling tensors:

$$J_{ij} = \frac{1}{N} \sum_{\alpha=1}^p \sum_{\beta=1}^p \xi_i^\alpha T_{\alpha\beta} \xi_j^\beta \quad (1)$$

where N denotes network dimension, p is the pattern count, $\xi_i^\alpha \in \{-1, +1\}$ is the i -th neuron value in pattern α , and $T_{\alpha\beta}$ represents the macroscopic pattern transition probability:

$$T_{\alpha\beta} = \mathbf{P}(\text{State}(t+1) = \xi^\alpha \mid \text{State}(t) = \xi^\beta) \quad (2)$$

- **Auto-Associative Fixed Points:** $T_{\alpha\beta} = \delta_{\alpha\beta}$ (Kronecker delta), recovering static Hebbian attractors.
- **Deterministic Sequence Chains:** $T_{\alpha\beta} = \delta_{\alpha, \beta+1}$, driving cyclical transitions $\xi^1 \rightarrow \xi^2 \rightarrow \dots \rightarrow \xi^p \rightarrow \xi^1$.
- **Functional Transition Kernels:** As noted (“could alter like a function”), $T_{\alpha\beta}$ can be any general Markovian probability transition function $T(\alpha, \beta)$.

2 Derivation of Macroscopic Joint Transition Probability (Step C)

We address the mathematical derivation of the microscopic-to-macroscopic joint probability product across all N neurons.

2.1 Microscopic Parallel Spin Dynamics & Factorization

In Little–Peretto parallel network dynamics, all N spin neurons $S_i \in \{-1, +1\}$ update simultaneously at discrete time steps $t \rightarrow t+1$. Because each neuron’s update at $t+1$ depends strictly on the global state $\mathbf{S}(t)$, individual neuron stochastic fluctuations are conditionally independent given $\mathbf{S}(t)$.

Consequently, the microscopic joint transition probability for state vector $\mathbf{S}(t+1) = \mathbf{s}'$ given $\mathbf{S}(t) = \mathbf{s}$ factorizes into a product over all N units:

$$\mathbf{P}(\mathbf{S}(t+1) = \mathbf{s}' \mid \mathbf{S}(t) = \mathbf{s}) = \prod_{i=1}^N \mathbf{P}(S_i(t+1) = s'_i \mid \mathbf{S}(t) = \mathbf{s}) \quad (3)$$

2.2 Glauber Thermal Activation Law

Under heat-bath thermal noise with inverse temperature $\beta = 1/T$, single-neuron transition probabilities follow the logistic distribution:

$$\mathbf{P}(S_i(t+1) = s'_i \mid \mathbf{S}(t) = \mathbf{s}) = \frac{1}{2} [1 + s'_i \tanh(\beta h_i(\mathbf{s}))] \quad (4)$$

where $h_i(\mathbf{s})$ is the local synaptic field acting on neuron i :

$$h_i(\mathbf{s}) = \sum_{j=1}^N J_{ij} S_j(t) \quad (5)$$

2.3 Field Evaluation at Source Pattern ξ^γ

Evaluating the local field when the network resides at source memory $\mathbf{S}(t) = \xi^\gamma$ and substituting Equation (1):

$$h_i(\xi^\gamma) = \sum_{j=1}^N \left[\frac{1}{N} \sum_{\alpha, \beta} \xi_i^\alpha T_{\alpha\beta} \xi_j^\beta \right] \xi_j^\gamma = \sum_{\alpha, \beta} \xi_i^\alpha T_{\alpha\beta} \left[\frac{1}{N} \sum_{j=1}^N \xi_j^\beta \xi_j^\gamma \right] \quad (6)$$

Assuming orthogonal memory vectors ($\frac{1}{N} \sum_j \xi_j^\beta \xi_j^\gamma = \delta_{\beta\gamma}$), Equation (6) collapses to:

$$h_i(\xi^\gamma) = \sum_{\alpha=1}^p \xi_i^\alpha T_{\alpha\gamma} \quad (7)$$

2.4 Full Macroscopic Transition Probability Formula

Setting target state $s'_i = \xi_i^\delta$ for all $i = 1, \dots, N$ yields the exact joint transition probability from pattern ξ^γ to pattern ξ^δ :

$$\mathbf{P}(\mathbf{S}(t+1) = \xi^\delta \mid \mathbf{S}(t) = \xi^\gamma) = \prod_{i=1}^N \frac{1}{2} \left[1 + \xi_i^\delta \tanh \left(\beta \sum_{\alpha=1}^p \xi_i^\alpha T_{\alpha\gamma} \right) \right] \quad (8)$$

3 Extension to Continuous Modern Hopfield & DAM

In Modern Hopfield Networks and Dense Associative Memory (DAM), interaction potentials use continuous vectors $\mathbf{x} \in \mathbb{R}^d$ and softmax attention operations over pattern matrix $\mathbf{X} = [\xi^1, \dots, \xi^p]$.

The single-step temporal retrieval update law is given by:

$$\mathbf{x}(t+1) = \mathbf{X} \mathbf{T}_{\text{seq}} \cdot \text{softmax}(\beta_{\text{hop}} \mathbf{X}^T \mathbf{x}(t)) \quad (9)$$

And the continuous pattern transition matrix $\mathbf{P} \in \mathbb{R}^{p \times p}$ reduces to the compact matrix equation:

$$\mathbf{P} = \mathbf{T}_{\text{seq}} \cdot \text{softmax}(\beta_{\text{hop}} \mathbf{X}^T \mathbf{X}) \quad (10)$$

4 Model Comparison Summary

Model Architecture	State Space	Synaptic Coupling / Energy	Capacity
Classical Hopfield	Binary spin $S_i \in \{-1, +1\}$	Bilinear $J_{ij} = \frac{1}{N} \sum \xi_i^\alpha T_{\alpha\beta} \xi_j^\beta$	$p \approx 0.14N$
Dense Assoc. Memory	Continuous $\mathbf{x} \in \mathbb{R}^d$	Polynomial $F(\mathbf{x}) = \mathbf{x}^n$	$p \propto N^{n-1}$
Modern Exp. Hopfield	Continuous vectors $\mathbf{x} \in \mathbb{R}^d$	Exponential $F(\mathbf{x}) = \exp(\beta \mathbf{x})$	$p \propto 2^{N/2}$

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